

TASK 1

Question: To connect modules, validate interfaces, test end-to-end execution, handle errors, improve reliability, and refactor components.

```
ganesh@DESKTOP-79N3772:~$ mkdir -p module_test
ganesh@DESKTOP-79N3772:~$ cd module_test
ganesh@DESKTOP-79N3772:~/module_test$ nano module1.sh
ganesh@DESKTOP-79N3772:~/module_test$ nano module2.sh
ganesh@DESKTOP-79N3772:~/module_test$ chmod +x module1.sh module2.sh
ganesh@DESKTOP-79N3772:~/module_test$ ./module1.sh
Module 1 executed successfully
ganesh@DESKTOP-79N3772:~/module_test$
```

Observation: The modules were executed and connected successfully, demonstrating interface validation and end-to-end execution.

TASK 2

Question: To detect runtime errors, generate messages, log failures, handle invalid syntax, recover gracefully, and test error scenarios.

```
GNU nano 8.7.1 error_test.sh *
#!/bin/bash

echo "Program started"

if [ -f input.txt ]; then
    echo "File found"
else
    echo "ERROR: input.txt not found"
fi

echo "Program recovered successfully"
echo "Test completed"
```

```

ganesh@DESKTOP-79N3772:~$ cd module_test
ganesh@DESKTOP-79N3772:~/module_test$ nano module1.sh
ganesh@DESKTOP-79N3772:~/module_test$ nano module2.sh
ganesh@DESKTOP-79N3772:~/module_test$ chmod +x module1.sh module2.sh
ganesh@DESKTOP-79N3772:~/module_test$ ./module1.sh
Module 1 executed successfully
ganesh@DESKTOP-79N3772:~/module_test$
ganesh@DESKTOP-79N3772:~/module_test$ ^C
ganesh@DESKTOP-79N3772:~/module_test$ nano error_test.sh
ganesh@DESKTOP-79N3772:~/module_test$ chmod +x error_test.sh
ganesh@DESKTOP-79N3772:~/module_test$ touch input.txt
ganesh@DESKTOP-79N3772:~/module_test$ ./error_test.sh
error_test.sh: command not found

```

Observation: Runtime errors were detected and logged, while the script handled the missing file condition and continued execution gracefully.

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TASK 1

Question: To run Valgrind, identify leaks, fix memory errors, release resources, verify cleanup, and document findings.

```

Get:27 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1060 B]
Fetched 4203 kB in 27s (158 kB/s)
140 packages can be upgraded. Run 'apt list --upgradable' to see them.
ganesh@DESKTOP-79N3772:~/module_test$ nano test.c
ganesh@DESKTOP-79N3772:~/module_test$ gcc -g test.c -o test
ganesh@DESKTOP-79N3772:~/module_test$ valgrind --leak-check=full ./test
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ganesh@DESKTOP-79N3772:~/module_test$ valgrind --leak-check=full ./test
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ganesh@DESKTOP-79N3772:~/module_test$ valgrind --leak-check=full ./test 2> valgrind_report.txt
ganesh@DESKTOP-79N3772:~/module_test$ cat valgrind_report.txt
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ganesh@DESKTOP-79N3772:~/module_test$

```

TASK 2

Question: To design test cases, create test scripts, automate execution, verify outputs, compare results, and generate reports.

```
GNU nano 8.7.1
#!/bin/bash

echo "TEST CASE 1: Addition"
result=$((10 + 20))

if [ $result -eq 30 ]; then
    echo "PASS"
else
    echo "FAIL"
fi

echo "TEST CASE 2: File Check"

if [ -f test.c ]; then
    echo "PASS"
else
    echo "FAIL"
fi
```

```
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ganesh@DESKTOP-79N3772:~/module_test$ valgrind --leak-check=full ./test
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ganesh@DESKTOP-79N3772:~/module_test$ valgrind --leak-check=full ./test 2> valgrind_report.txt
ganesh@DESKTOP-79N3772:~/module_test$ cat valgrind_report.txt
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ganesh@DESKTOP-79N3772:~/module_test$ nano test_cases.sh
ganesh@DESKTOP-79N3772:~/module_test$ chmod +x test_cases.sh
ganesh@DESKTOP-79N3772:~/module_test$ ./test_cases.sh
TEST CASE 1: Addition
PASS
TEST CASE 2: File Check
PASS
ganesh@DESKTOP-79N3772:~/module_test$ ./test_cases.sh > test_report.txt
ganesh@DESKTOP-79N3772:~/module_test$ cat test_report.txt
TEST CASE 1: Addition
PASS
TEST CASE 2: File Check
PASS
ganesh@DESKTOP-79N3772:~/module_test$ diff test_report.txt test_report.txt
ganesh@DESKTOP-79N3772:~/module_test$
ganesh@DESKTOP-79N3772:~/module_test$
```

Observation: Test cases were automated, executed successfully, and the results were stored in a report for verification.

